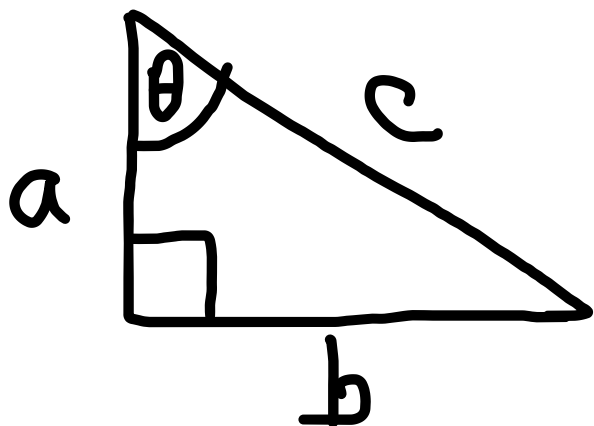


Trigonometric functions:



$$\sin \theta = \frac{b}{c}$$

$$\cos \theta = \frac{a}{c}$$

$$\tan \theta = \frac{b}{a}$$

Useful: SOH, CAH, TOA

$$\sin = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\cos = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

$$\tan = \frac{\text{Opposite}}{\text{Adjacent}}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

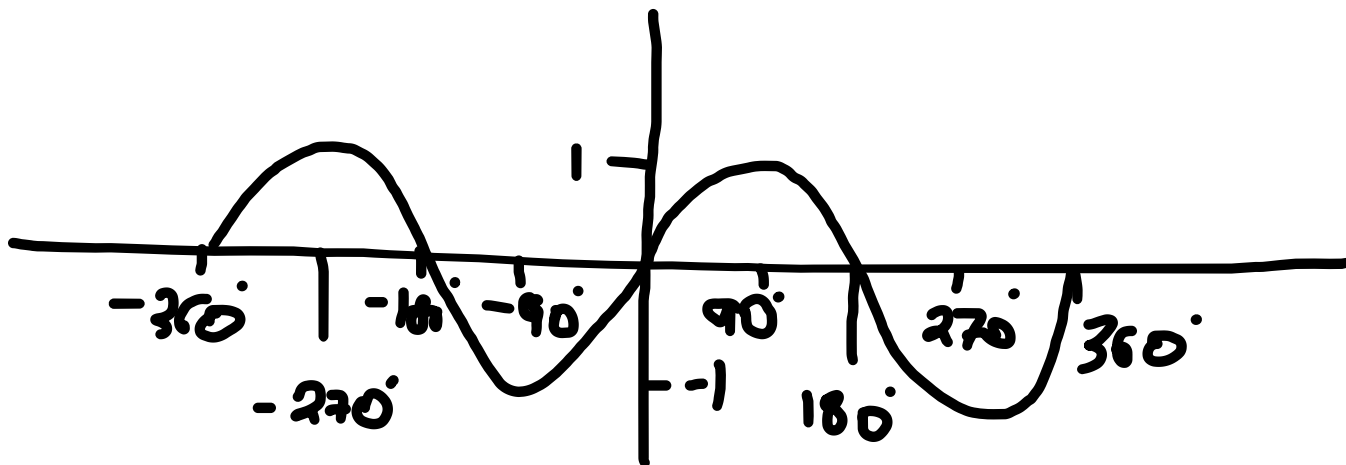
$$\sec \theta = \frac{1}{\cos \theta}$$

$$\csc \theta = \frac{1}{\sin \theta}$$

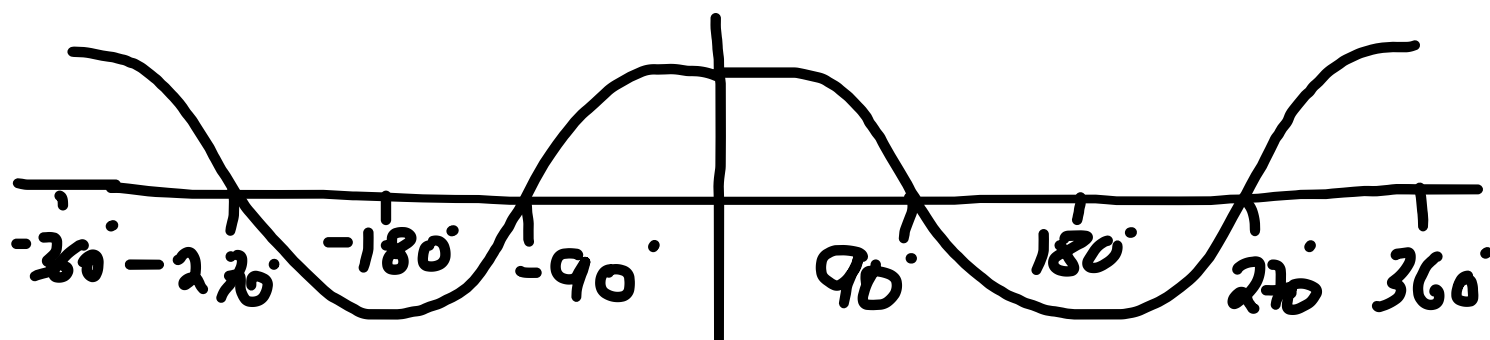
$$\cot \theta = \frac{1}{\tan \theta}$$

Graphs of trigonometric functions:

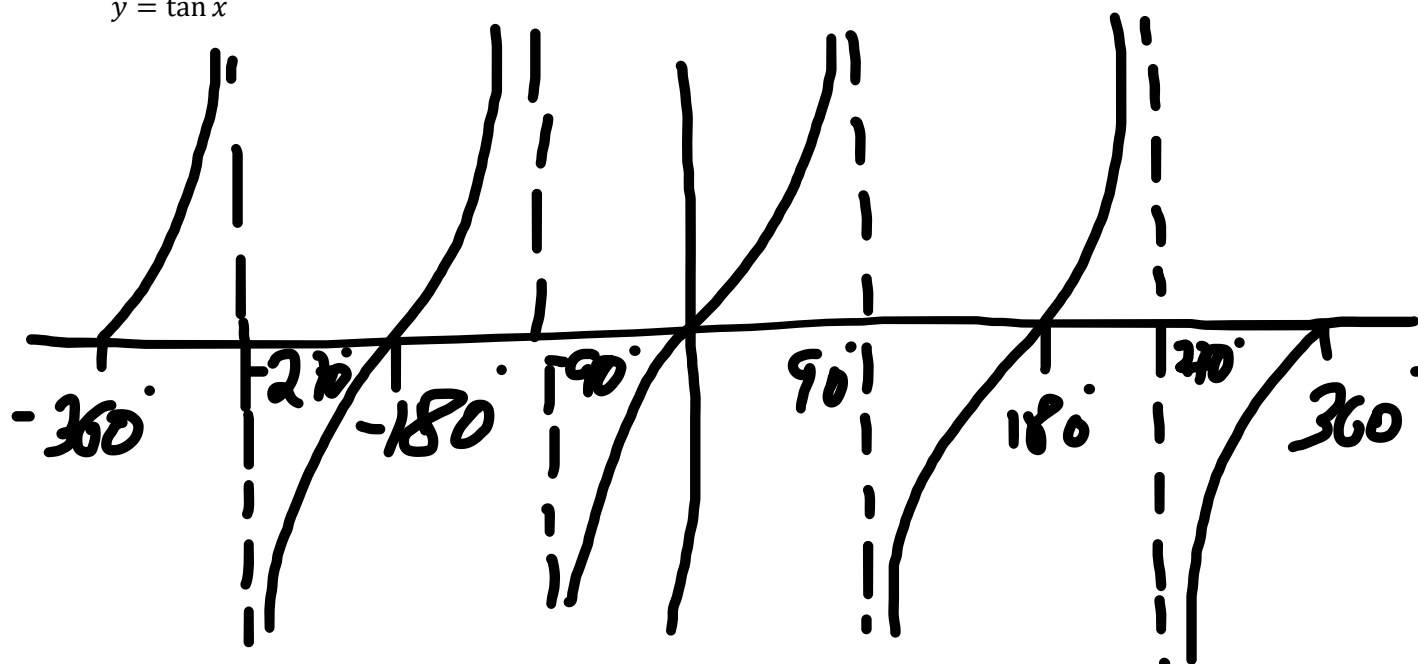
$$y = \sin x$$



$$y = \cos x$$



$$y = \tan x$$



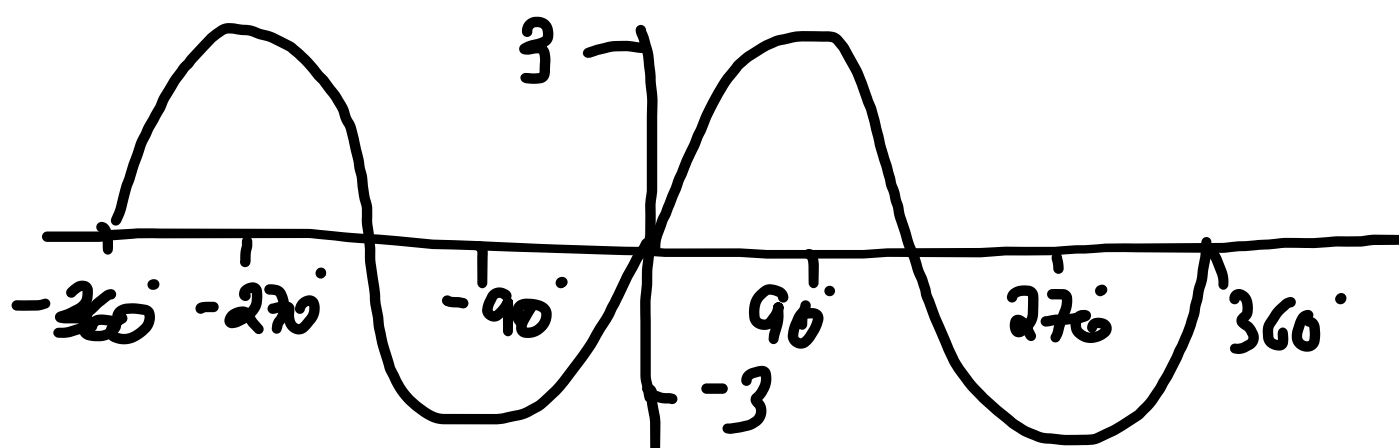
$$y = a \sin bx + c$$

a = amplitude

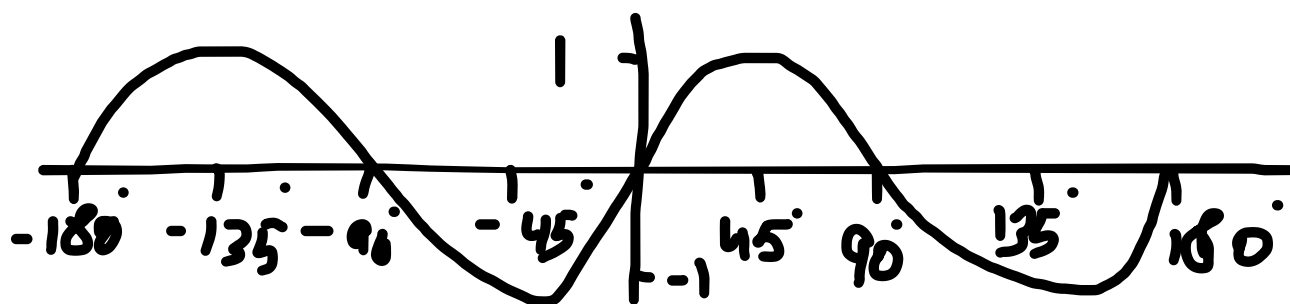
$$\frac{360^\circ}{b} = \text{period}$$

c = y-intercept

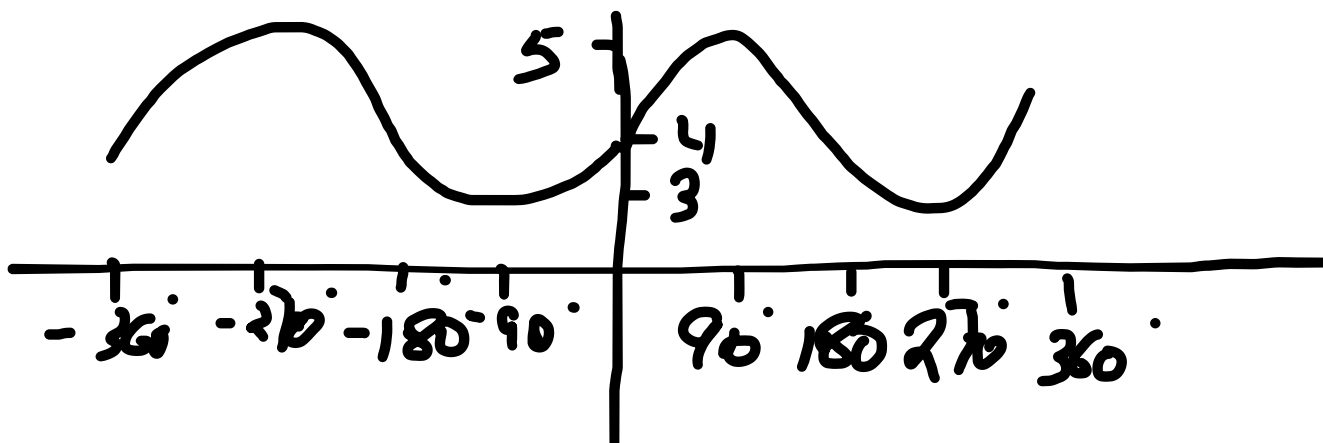
$$y = 3 \sin x$$



$$y = \sin 2x$$



$$y = \sin x + 4$$



$$y = a \cos bx + c$$

a = amplitude

$$\frac{360^\circ}{b} = \text{period}$$

c = y-intercept

Behaves similar to $y = a \sin bx + c$

$$y = a \tan bx + c$$

Amplitude does not exist, a affects the gradient only

$$\frac{180^\circ}{b} = \text{period}$$

c = y-intercept

Behaves similar to $y = a \sin bx + c$ except for the amplitude

Has asymptotes, originally at 90° and 270° , repeating periodically

Relationships:

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\csc^2 A = 1 + \cot^2 A$$

Solving trigonometric equations:

$$\tan \theta = \tan(\theta + 180)$$

$$\tan \theta = \tan(\theta - 180)$$

$$\sin \theta = \sin(\theta + 360)$$

$$\sin \theta = \sin(180 - \theta)$$

$$\sin \theta = -\sin -\theta$$

$$\cos \theta = \cos(\theta + 360)$$

$$\cos \theta = \cos(360 - \theta)$$

$$\cos \theta = \cos -\theta$$

Q1. $4 \cot \theta = \tan \theta$ for $-360^\circ < \theta < 360^\circ$

$$\frac{4}{\tan \theta} = \tan \theta$$

$$4 = \tan^2 \theta$$

$$\tan \theta = \pm 2$$

$$\theta = \tan^{-1} 2 = 63.43^\circ$$

$$\theta = \tan^{-1} -2 = -63.43^\circ$$

$$\theta = -243.4^\circ, -63.4^\circ, 63.4^\circ, 116.6^\circ, 243.4^\circ, 296.6^\circ$$

Q2. $2 \sec^2 \theta + \tan \theta - 3 = 0$ for $0^\circ < \theta < 360^\circ$

$$2(1 + \tan^2 \theta) + \tan \theta - 3 = 0$$

$$2 + 2 \tan^2 \theta + \tan \theta - 3 = 0$$

$$2 \tan^2 \theta + \tan \theta - 1 = 0$$

$$\tan \theta = \frac{1}{2}, -1$$

$$\theta = \tan^{-1} \frac{1}{2} = 26.57^\circ$$

$$\theta = \tan^{-1} -1 = -45^\circ$$

$$\theta = 26.6^\circ, 135^\circ, 206.6^\circ, 315^\circ$$

Q3. $5 \sin \frac{\theta}{3} + 2 \cos \frac{\theta}{3} = 0$ for $-\pi < \theta < \pi$

$$5 \tan \frac{\theta}{3} + 2 = 0$$

$$\tan \frac{\theta}{3} = \frac{-2}{5}$$

$$\frac{\theta}{3} = \tan^{-1} \frac{-2}{5} = -0.3805$$

$$\frac{\theta}{3} = 2.761$$

$$\theta = -1.142$$

Q4. $3 \csc \left(2\theta - \frac{\pi}{12} \right) = 4$ for $0 < \theta < \pi$

$$\frac{3}{\sin \left(2\theta - \frac{\pi}{12} \right)} = 4$$

$$\sin \left(2\theta - \frac{\pi}{12} \right) = \frac{3}{4}$$

$$2\theta - \frac{\pi}{12} = 0.8481, 7.131, 2.293$$

$$\theta = 0.555, 1.28$$

Proving trigonometric relationships:

Q1. Prove that $\sin x \tan x + \cos x = \sec x$

$$LHS = \frac{\sin^2 x}{\cos x} + \cos x = \frac{\sin^2 x + \cos^2 x}{\cos x} = \frac{1}{\cos x} = \sec x = RHS$$

Q2. Prove that $\frac{\sin \theta}{1 + \cos \theta} + \frac{1 + \cos \theta}{\sin \theta} = 2 \csc \theta$

$$\begin{aligned} LHS &= \frac{\sin^2 \theta + (1 + \cos \theta)^2}{\sin \theta (1 + \cos \theta)} = \frac{\sin^2 \theta + 1 + 2 \cos \theta + \cos^2 \theta}{\sin \theta + \sin \theta \cos \theta} = \frac{2 + 2 \cos \theta}{\sin \theta (1 + \cos \theta)} \\ &= \frac{2(1 + \cos \theta)}{\sin \theta (1 + \cos \theta)} = \frac{2}{\sin \theta} = 2 \csc \theta = RHS \end{aligned}$$